FET DC bias

Answers and worked solutions · Day 17 · Mon 9 Nov · pack E10

5. Practice

From your course files

WS8 Q2 (course question).

> **GIVEN:** $V_{DD} = 18$ V, $R_D = 2$ kΩ, $R_G = 2$ MΩ, $V_{GG} = 4$ V with its + terminal at ground, $I_{DSS} = 8$ mA, $V_P = -4$ V.
>
> $V_{GS} = -V_{GG} = -4\text{ V} = V_P$. So $I_D = 8(1-1)^2 = 0$ mA. The JFET is pinched off (cut off).
>
> $V_D = 18 - 0(2) = 18$ V.
>
> **ANSWER:** $V_{GS} = -4$ V, $V_D = 18$ V. The point of this question is to recognise $V_{GS} = V_P$ and avoid computing anything further.

WS8 Q1 (course question).

> **Given (from figure):** $V_{DD} = 14$ V, $R_D = 1.8$ kΩ, $R_G = 1$ MΩ, $V_{GG} = 1.5$ V with its + terminal at ground, $I_{DSS} = 12$ mA, $V_P = -4$ V.
>
> **(a) Transfer curve points:**
>
> | V_{GS} (V) | 0 | -1.2 ($0.3V_P$) | -2 ($0.5V_P$) | -4 |
> |---|---|---|---|---|
> | I_D (mA) | 12 | 6 | 3 | 0 |
>
> Draw a smooth parabola through these points.
>
> **(b) Graphical:** $V_{GS} = -V_{GG} = -1.5$ V. This is a vertical line. It meets the curve at about $I_{DQ} \approx 4.7$ mA. Then $V_{DSQ} = 14 - 4.7(1.8) \approx 5.5$ V.
>
> **(c) Shockley:**
>
> $$I_{DQ} = 12\left(1 - \frac{-1.5}{-4}\right)^2 = 12(0.625)^2 = 4.69\text{ mA}$$
>
> $$V_{DSQ} = 14 - (4.69)(1.8) = 5.56\text{ V}$$
>
> The two methods agree to within graph-reading accuracy.
>
> **Saturation check:** $V_{GS} - V_P = 2.5$ V, and $5.56 \geq 2.5$. Saturated.

ANSWER: $I_{DQ} = 4.69$ mA, $V_{DSQ} = 5.56$ V.

WS8 Q3 (course question).

GIVEN: $V_{DD} = 18$ V, $R_D = 1.5$ kΩ, $R_S = 750$ Ω, $R_G = 1$ MΩ to ground, $I_{DSS} = 10$ mA, $V_P = -4$ V.

Bias line: $V_G = 0$, so $V_{GS} = -0.75 I_D$ (I_D in mA).

Substitute:

$$I_D = 10\left(1 - \frac{-0.75 I_D}{-4}\right)^2 = 10\,(1 - 0.1875 I_D)^2$$

$$0.3516\, I_D^2 - 4.75\, I_D + 10 = 0$$

The roots are:

- $I_D = 10.90$ mA, which gives $V_{GS} = -8.18$ V. This is below $V_P = -4$ V, so reject it.
- $I_D = 2.61$ mA, which gives $V_{GS} = -1.96$ V. This is valid.

ANSWER: $I_{DQ} = 2.61$ mA (with $V_{GSQ} = -1.96$ V).

As an extra check, $V_{DS} = 18 - 2.61(2.25) = 12.1$ V, which is at least $V_{GS} - V_P = 2.04$ V, so it is saturated. The figure in Section 3 shows this crossing.

WS8 Q4 (course question).

GIVEN: $V_{DD} = 18$ V, $R_D = 2$ kΩ, $R_S = 0.51$ kΩ, $R_G = 1$ MΩ to ground, $V_P = -4$ V, measured $V_S = 1.7$ V.

a. $I_{DQ} = V_S / R_S = 1.7 / 0.51 = 3.33$ mA.

b. $V_G = 0$, so $V_{GSQ} = V_G - V_S = -1.7$ V.

c. Rearrange Shockley:

$$I_{DSS} = \frac{I_D}{(1 - V_{GS}/V_P)^2} = \frac{3.333}{(1 - 0.425)^2} = \frac{3.333}{0.3306} = 10.08\ \text{mA}$$

d. $V_D = 18 - 3.333(2) = 11.33$ V.

e. $V_{DS} = V_D - V_S = 11.33 - 1.7 = 9.63$ V.

Saturation: $V_{GS} - V_P = 2.3$ V, and $9.63 \geq 2.3$. Saturated.

ANSWERS: 3.33 mA, -1.7 V, 10.1 mA, 11.33 V, 9.63 V.

WS9 Q1 (course question).

> **READING THE FIGURE:** the gate (the terminal with the arrow) is grounded. The left channel terminal goes through 2.2 kΩ to +14 V, so that is the drain. The right terminal goes through 0.39 kΩ to ground, and that is the source, where V_S is measured. $I_{DSS} = 6$ mA, $V_P = -6$ V.
>
> Bias line: $V_G = 0$, so $V_{GS} = -0.39 I_D$.
>
> Substitute: $I_D = 6(1 - 0.065 I_D)^2$. The roots are:
>
> - 66.7 mA, which gives $V_{GS} = -26$ V. Reject it.
> - 3.55 mA, which gives $V_{GS} = -1.38$ V. This is valid.
>
> $V_S = I_D R_S = 3.55(0.39) = 1.38$ V.
>
> Check: $V_D = 14 - 3.55(2.2) = 6.19$ V and $V_{DS} = 4.80$ V. Also $V_{GS} - V_P = 4.62$ V, so it is saturated, but only by a small margin.
>
> **ANSWER:** $V_S = 1.38$ V.

WS9 Q3 (course question).

> **READING THE FIGURE:** the gate is grounded. $R_D = 3$ kΩ goes to +20 V. $R_S = 1.2$ kΩ goes to **+2 V**, which is positive. $V_{DS} = 4$ V is given. No device parameters are needed.
>
> Output KVL: $V_{DS} = 20 - 2 - I_D(3 + 1.2)$, so $4 = 18 - 4.2 I_D$.
>
> $I_D = 14/4.2 = 3.33$ mA.
>
> $V_D = 20 - 3.33(3) = 10$ V. $V_S = 2 + 3.33(1.2) = 6$ V. Check: $V_{DS} = 10 - 6 = 4$ V. This matches.
>
> $V_{GS} = V_G - V_S = 0 - 6 = -6$ V.
>
> **ANSWERS:** $I_D = 3.33$ mA, $V_D = 10$ V, $V_S = 6$ V, $V_{GS} = -6$ V. Because the source supply is positive, it adds to the source voltage instead of subtracting.

WS9 Q2 (course question).

> **READING THE FIGURE:** the gate is grounded. The drain goes through 2.2 kΩ to +16 V. The source goes through 2.2 kΩ to −4 V. $I_{DSS} = 6$ mA, $V_P = -6$ V. The figure labels the gate–source voltage "V_{DSQ}". This is a typo for V_{GSQ}, because the + sign is at the gate.
>
> Source voltage: $V_S = -4 + 2.2 I_D$. So $V_{GS} = 0 - V_S = 4 - 2.2 I_D$.

Substitute: $I_D = 6\left(1 + \dfrac{4 - 2.2I_D}{6}\right)^2 = 6\,(1.667 - 0.367I_D)^2$. The roots are:

- 7.62 mA, which gives $V_{GS} = -12.8$ V. Reject it.
- 2.71 mA, which gives $V_{GS} = -1.97$ V. This is valid.

a. $I_{DQ} = 2.71$ **mA,** $V_{GSQ} = -1.97$ **V.**

b. $V_S = -4 + 2.71(2.2) = +1.97$ V. $V_D = 16 - 2.71(2.2) = 10.03$ V.
$V_{DS} = 10.03 - 1.97 = 8.07$ V.

Using the formula from the slides:
$V_{DS} = V_{DD} + V_{SS} - I_D(R_D + R_S) = 20 - 2.71(4.4) = 8.07$ V. This agrees.

Saturation: $V_{GS} - V_P = 4.03$ V, and $8.07 \geq 4.03$. Saturated.

ANSWERS: 2.71 mA, -1.97 V, $V_{DS} = 8.07$ V, $V_S = 1.97$ V.

WS8 Q5 (course question).

GIVEN: $V_{DD} = 16$ V, $R_1 = 36$ kΩ, $R_2 = 12$ kΩ, $R_D = 2$ kΩ, $I_{DSS} = 12$ mA, $V_P = -8$ V. The target is $V_D = 10$ V.

1. $I_D = (16 - 10)/2 = 3$ mA.
2. Invert Shockley: $V_{GS} = V_P(1 - \sqrt{I_D/I_{DSS}}) = -8(1 - \sqrt{0.25}) = -4$ V.
3. $V_G = 12(16)/(36 + 12) = 4$ V.
4. $V_S = V_G - V_{GS} = 4 - (-4) = 8$ V.
5. $R_S = V_S/I_D = 8/3 = 2.67$ kΩ.

Answer (standard method): $R_S = 2.67$ kΩ.

Saturation check, which the question does not ask for: $V_{DS} = 10 - 8 = 2$ V, but $V_{GS} - V_P = -4 + 8 = 4$ V. Since $2 < 4$, the JFET would actually be in the **ohmic region** at this Q-point, so Shockley's equation does not strictly apply. The expected workshop answer is still 2.67 kΩ. I checked with an ohmic-region (triode) model, which is not part of this course: with $R_S = 2.67$ kΩ, the real circuit gives $I_D \approx 2.89$ mA and $V_D \approx 10.2$ V, so the design is still close. In an exam, give 2.67 kΩ and add one line noting the saturation check fails. That shows the examiner you know the limits of the model.

WS8 Q6 (course question).

GIVEN: $V_{DD} = 24$ V, $R_1 = 10$ MΩ, $R_2 = 6.8$ MΩ, $R_D = 2.2$ kΩ, $R_S = 0.75$ kΩ, $V_{GS(\text{Th})} = 3$ V, $I_{D(\text{on})} = 5$ mA at $V_{GS(\text{on})} = 6$ V.

1. $k = 5/(6 - 3)^2 = 0.556$ mA/V².
2. $V_G = 6.8(24)/16.8 = 9.71$ V. Bias line: $V_{GS} = 9.71 - 0.75I_D$.

3. Substitute: $I_D = 0.556\,(9.71 - 0.75I_D - 3)^2 = 0.556\,(6.71 - 0.75I_D)^2$.
4. The roots are:
   - $I_D = 16.1$ mA, which gives $V_{GS} = -2.39$ V. This is below V_T, so reject it.
   - $I_D = 4.97$ mA, which gives $V_{GS} = 5.99$ V. This is valid.

a. $I_{DQ} = 4.97$ **mA,** $V_{GSQ} = 5.99$ **V.** The device is running almost exactly at its $I_{D(\text{on})}$ point.

b. $V_D = 24 - 4.97(2.2) = 13.07$ V. $V_S = 4.97(0.75) = 3.72$ V.

Saturation: $V_{DS} = 9.35$ V, and $V_{GS} - V_T = 2.99$ V, so it is saturated.

ANSWERS: 4.97 mA, 5.99 V, $V_D = 13.1$ V, $V_S = 3.72$ V.

WS9 Q4 (course question).

READING THE FIGURE: this is an n-channel D-MOSFET (continuous channel bar). The gate is grounded. $R_D = 2.2$ kΩ goes to +18 V. $R_S = 0.39$ kΩ goes to −4 V. $I_{DSS} = 8$ mA, $V_P = -8$ V.

Standard method:

$V_S = -4 + 0.39I_D$, so $V_{GS} = 4 - 0.39I_D$.

Substitute: $I_D = 8\left(1 + \dfrac{4 - 0.39I_D}{8}\right)^2 = 8(1.5 - 0.04875I_D)^2$. The roots are:

- 105 mA, which gives $V_{GS} = -37$ V. Reject it.
- 9.01 mA, which gives $V_{GS} = +0.49$ V. This is allowed for a D-MOSFET.

a. $I_{DQ} = 9.01$ mA, $V_{GSQ} = +0.49$ V.

b. $V_S = -4 + 9.01(0.39) = -0.49$ V. $V_D = 18 - 9.01(2.2) = -1.81$ V. So $V_{DS} = V_D - V_S = -1.32$ V.

This result is impossible. V_{DS} cannot be negative for an n-channel device biased this way. The largest current the resistors can pass is $(18 + 4)/(2.2 + 0.39) = 8.49$ mA, which happens when $V_{DS} = 0$. That is less than the 9.01 mA that Shockley's equation predicts. So the saturation check fails, and the MOSFET is in the ohmic region.

An ohmic-region (triode) model, which is not part of this course, gives about $I_D \approx 7.0$ mA, $V_{GS} \approx +1.26$ V, $V_S \approx -1.26$ V and $V_{DS} \approx 3.8$ V. You cannot do this calculation with this course's tools.

What to write in an exam: give the Shockley values for (a). For (b), state $V_{DS} = -1.3$ V, then add that this violates $V_{DS} \geq V_{GS} - V_P$, so the device is not in saturation and the Shockley Q-point is not valid. I believe this is the textbook problem the question is taken

from. Its answer is usually quoted without this check, so the intended answer is probably $I_D \approx 9$ mA, $V_{GS} \approx 0.5$ V, $V_{DS} \approx -1.3$ V, $V_S \approx -0.49$ V.

New questions

N1. [4 marks]

$I_D = 8(1 - 2/6)^2 = 8(0.6667)^2 = 3.56$ mA.

$V_{DS} = 15 - 3.56(1.5) = 9.67$ V.

Saturation: $V_{GS} - V_P = 4$ V, and $9.67 \geq 4$. Saturated.

ANSWER: $I_{DQ} = 3.56$ mA, $V_{DSQ} = 9.67$ V.

N2. [6 marks]

a. The bias line is $V_{GS} = -I_D(1\text{ k}) = -I_D$. Substitute: $I_D = 12(1 - 0.2I_D)^2$, which gives $0.48I_D^2 - 5.8I_D + 12 = 0$.

The roots are:

- 9.43 mA, which gives $V_{GS} = -9.43$ V. This is below $V_P = -5$ V, so reject it.
- 2.65 mA, which gives $V_{GS} = -2.65$ V. This is in $[-5, 0]$, so keep it.

$I_{DQ} = 2.65$ **mA,** $V_{GSQ} = -2.65$ **V.**

b. $V_{DS} = 20 - 2.65(2.2 + 1) = 11.5$ V. Also $V_{GS} - V_P = 2.35$ V, and $11.5 \geq 2.35$, so it is saturated.

N3. [6 marks] Design.

1. $V_G = 100(20)/1000 = 2$ V.
2. $V_{GS} = -4(1 - \sqrt{4/10}) = -4(1 - 0.6325) = -1.47$ V.
3. $V_S = V_G - V_{GS} = 2 + 1.47 = 3.47$ V, so $R_S = 3.47/4 = 0.868$ kΩ.
4. $R_D = (20 - 12)/4 = 2$ kΩ.
5. Check: $V_{DS} = 12 - 3.47 = 8.53$ V, and $V_{GS} - V_P = 2.53$ V. Since $8.53 \geq 2.53$, it is saturated.

ANSWER: $R_S \approx 868$ Ω (use the nearest standard value, 820 Ω or 910 Ω), $R_D = 2$ kΩ.

N4. [6 marks]

$V_G = 1(18)/9 = 2$ V. The bias line is $V_{GS} = 2 - 0.2I_D$.

Substitute: $I_D = 8\left(1 + \dfrac{2 - 0.2I_D}{4}\right)^2 = 8(1.5 - 0.05I_D)^2$.

The roots are:

- 101 mA, which gives $V_{GS} = -18.2$ V. Reject it.
- 8.90 mA, which gives $V_{GS} = +0.22$ V. This is valid.

$I_{DQ} = 8.90$ **mA,** $V_{GSQ} = +0.22$ **V.** V_{GSQ} is positive, so the D-MOSFET is in enhancement mode and $I_{DQ} > I_{DSS}$.

$V_D = 18 - 8.90(1.2) = 7.32$ V. $V_S = 8.90(0.2) = 1.78$ V. $V_{DS} = 5.54$ V.

Saturation: $V_{GS} - V_P = 4.22$ V, and $5.54 \geq 4.22$. Saturated.

N5. [8 marks] Chained.

a. $k = 4/(6-2)^2 = 0.25$ mA/V².

b. The bias line is $V_{GS} = 15 - 1.5I_D$. Substitute: $I_D = 0.25\,(13 - 1.5I_D)^2$. The roots are:

- 13.6 mA, which gives $V_{GS} = -5.37$ V. This is below V_T, so reject it.
- 5.53 mA, which gives $V_{GS} = 6.70$ V. This is valid.

$I_{DQ} = 5.53$ **mA,** $V_{GSQ} = 6.70$ **V.**

c. $V_{DSQ} = V_{GSQ} = 6.70$ V. No check is needed, because saturation needs $V_{DS} \geq V_{GS} - V_T$, and with $V_{DS} = V_{GS}$ this always holds.

d. For $V_{DS} = V_{GS} = 8$ V, $I_D = 0.25(8-2)^2 = 9$ mA. Then $R_D = (15-8)/9 = 0.778$ kΩ.

ANSWER: $R_D = 778\ \Omega$.

N6. [8 marks] Common-gate design, hard.

a. $V_{GS} = -5(1 - \sqrt{0.5}) = -1.46$ V. With the gate grounded, $V_S = -V_{GS} = +1.46$ V. Also $V_S = -10 + 5R_S$, so $R_S = (10 + 1.46)/5 = 2.29$ kΩ.

b. $R_D = (15-6)/5 = 1.8$ kΩ. $V_S = 1.46$ V, so $V_{DS} = 6 - 1.46 = 4.54$ V. Also $V_{GS} - V_P = 3.54$ V, and $4.54 \geq 3.54$, so it is saturated.

c. With $R_D = 2.2$ kΩ, $V_D = 15 - 11 = 4$ V and $V_{DS} = 2.54$ V, which is less than 3.54 V. **It is not saturated.**

The limit is $V_D \geq V_S + (V_{GS} - V_P) = 1.46 + 3.54 = 5$ V. So $R_D \leq (15-5)/5 = 2.0$ kΩ.

> Note that I_D does not depend on R_D while the JFET stays saturated, because R_D is not in the gate–source loop.

6. Quick recall check

Question

1. $V_{GS} = -I_D R_S$. It passes through the origin. Pick any I_D to get a second point.
2. Keep the root with $V_P \leq V_{GS} \leq 0$. Reject the one with $V_{GS} < V_P$.
3. Yes, for a D-MOSFET when $V_{GSQ} > 0$ (enhancement mode). It never happens for a JFET.
4. JFET: $V_{DS} \geq V_{GS} - V_P$. E-MOSFET: $V_{DS} \geq V_{GS} - V_{GS(\mathrm{Th})}$.
5. $V_{DS} = V_{GS}$, because $I_G = 0$ means no voltage drops across R_G, so $V_G = V_D$.